

Assignment ::- 4

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Subject	DBMS

Q1) Create a MySQL database named CollegeDB and create the following tables:

Student Table

Column Name	Data Type	Constraint
StudentID	INT	PRIMARY KEY
StudentName	VARCHAR(50)	
Column Name	Data Type	Constraint
City	VARCHAR(30)	

Course Table

Column Name	Data Type	Constraint
CourseID	INT	PRIMARY KEY
CourseName	VARCHAR(50)	
Fees	DECIMAL(10,2)	

Enrollment Table

Column Name	Data Type	Constraint
EnrollmentID	INT	PRIMARY KEY
StudentID	INT	FOREIGN KEY
CourseID	INT	FOREIGN KEY
EnrollmentDate	DATE	

CREATE DATABASE CollegeDB;

USE CollegeDB;

CREATE TABLE Student (

StudentID INT PRIMARY KEY,

StudentName VARCHAR(50),

City VARCHAR(30)

);

CREATE TABLE Course (

CourseID INT PRIMARY KEY,

CourseName VARCHAR(50),

Fees DECIMAL(10,2)

);

CREATE TABLE Enrollment (

EnrollmentID INT PRIMARY KEY,

StudentID INT,

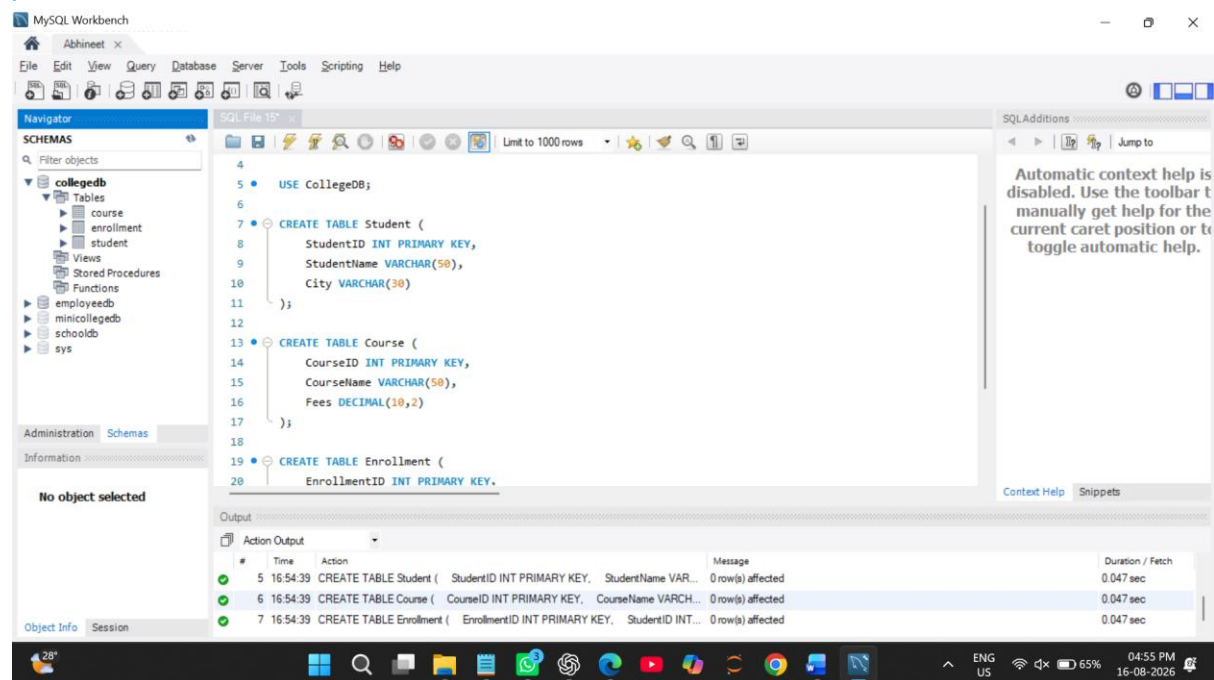
CourseID INT,

EnrollmentDate DATE,

FOREIGN KEY (StudentID) REFERENCES Student(StudentID),

FOREIGN KEY (CourseID) REFERENCES Course(CourseID)

);



Q2) Insert at least 6 records into the Student table, 4 records into the Course table, and 5 records into the Enrollment table.

Make sure:

Requirement

At least one student should not have enrollment

At least one course should not have any enrolled student

-- Insert records into Student table

```
INSERT INTO Student (StudentID, StudentName, City)
```

```
VALUES
```

```
(1, 'Rahul Sharma', 'Mumbai'),
```

```
(2, 'Priya Patil', 'Pune'),
```

```
(3, 'Aman Verma', 'Nashik'),
```

```
(4, 'Sneha Joshi', 'Mumbai'),
```

```
(5, 'Karan Mehta', 'Thane'),
```

```
(6, 'Neha Shah', 'Pune');
```

```
-- Insert records into Course table
```

```
INSERT INTO Course (CourseID, CourseName, Fees)
```

```
VALUES
```

```
(101, 'BSc IT', 45000.00),
```

```
(102, 'BCA', 40000.00),
```

```
(103, 'BSc CS', 42000.00),
```

```
(104, 'Data Science', 55000.00);
```

```
-- Insert records into Enrollment table
```

```
INSERT INTO Enrollment (EnrollmentID, StudentID, CourseID, EnrollmentDate)
```

```
VALUES
```

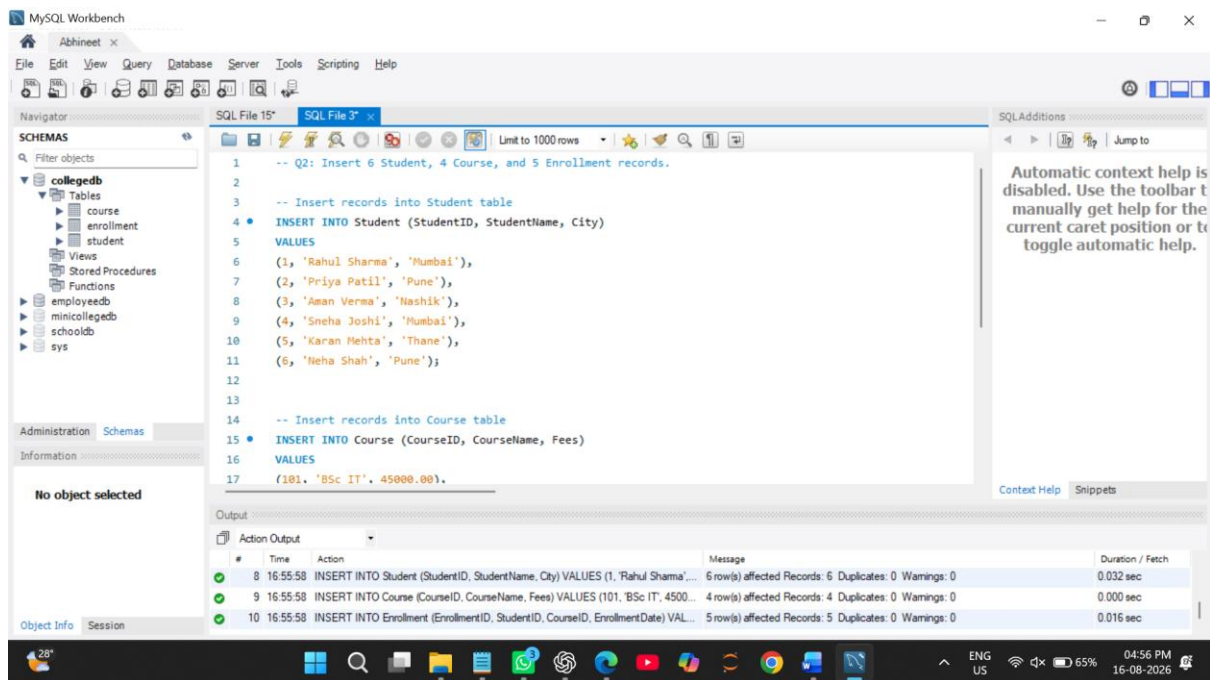
```
(1, 1, 101, '2026-07-01'),
```

```
(2, 2, 102, '2026-07-02'),
```

```
(3, 3, 103, '2026-07-03'),
```

```
(4, 4, 101, '2026-07-04'),
```

(5, 5, 102, '2026-07-05');



Q3) Write a MySQL query using INNER JOIN to display students who have enrolled in courses.

The output should include:

Column
StudentID
StudentName
CourseID
EnrollmentDate

SELECT

Student.StudentID,

Student.StudentName,

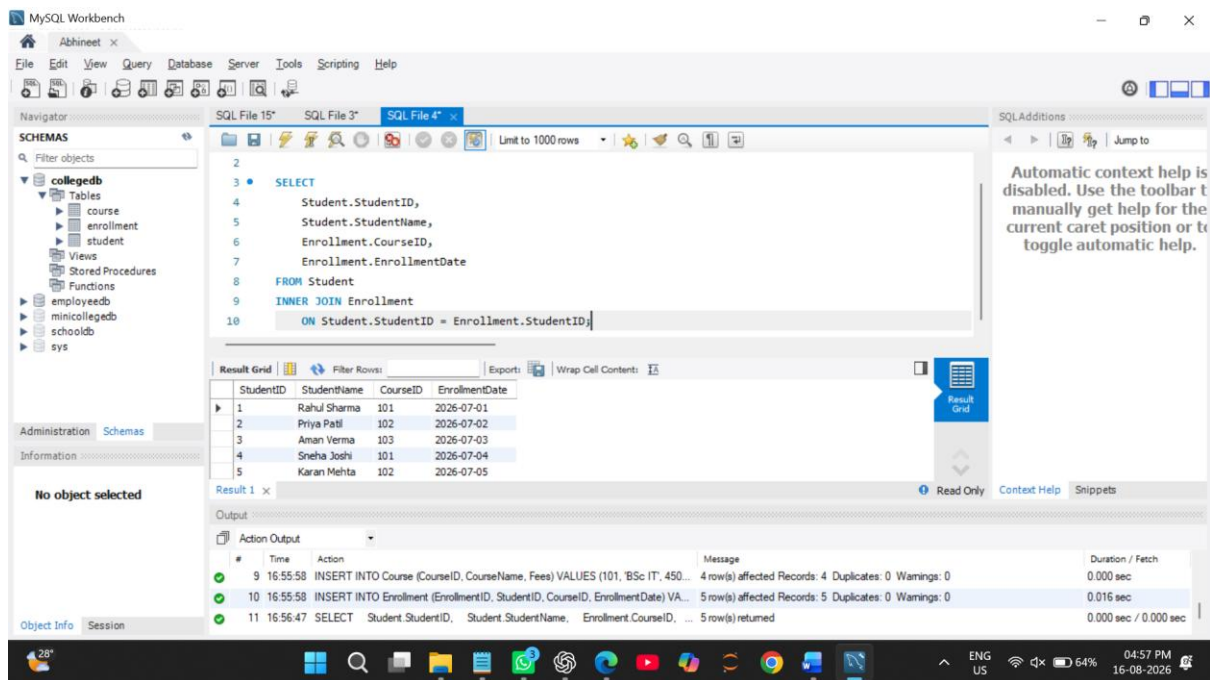
Enrollment.CourseID,

Enrollment.EnrollmentDate

FROM Student

INNER JOIN Enrollment

ON Student.StudentID = Enrollment.StudentID;



Q4) Write a MySQL query to display student name, city, course name, fees, and enrollment date using multiple tables.

Use:

Tables Required
Student
Course
Enrollment

SELECT

Student.StudentName,

Student.City,

Course.CourseName,

Course.Fees,

Enrollment.EnrollmentDate

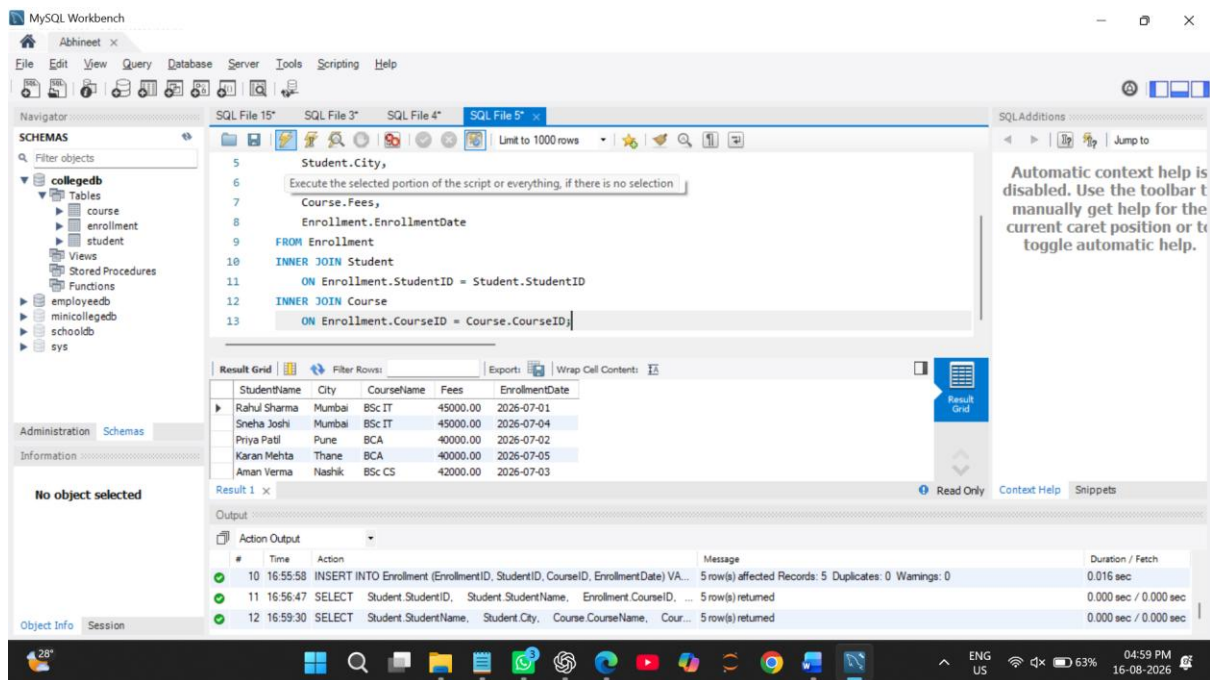
FROM Enrollment

INNER JOIN Student

ON Enrollment.StudentID = Student.StudentID

INNER JOIN Course

ON Enrollment.CourseID = Course.CourseID;



Q5) Write a MySQL query using LEFT JOIN to display all students, including those who have not enrolled in any course.

The output should include:

Column
StudentID
StudentName
City

SELECT

Student.StudentID,

Student.StudentName,

Student.City,

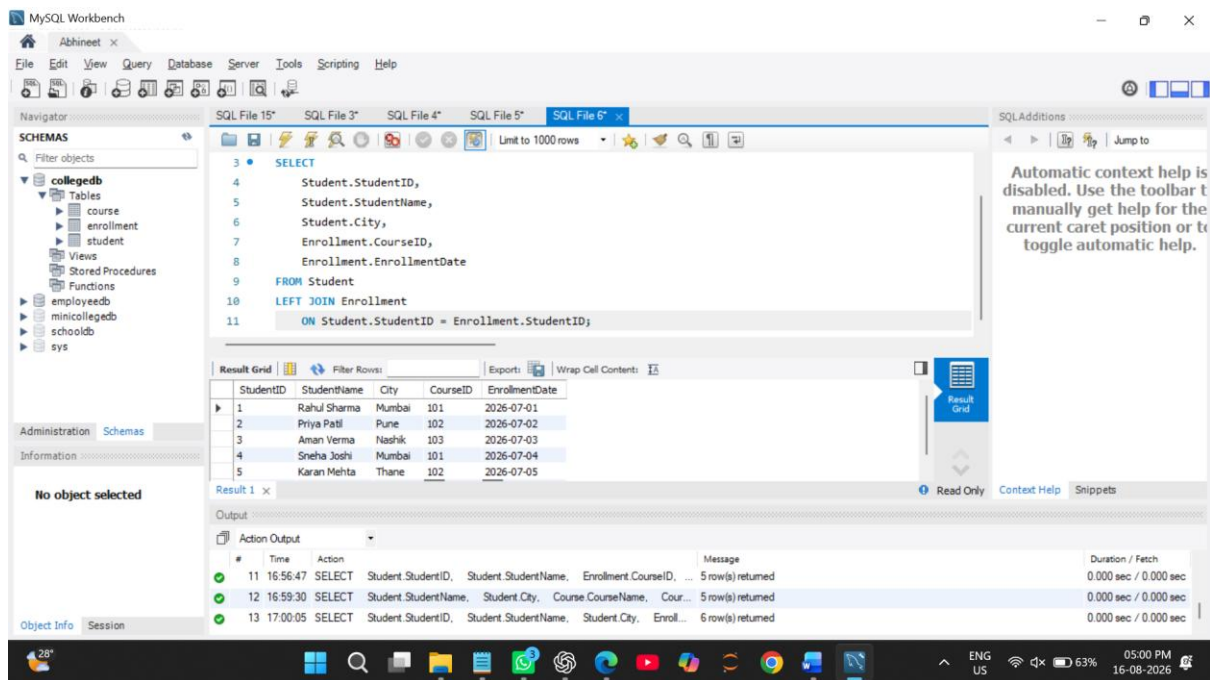
Enrollment.CourseID,

Enrollment.EnrollmentDate

FROM Student

LEFT JOIN Enrollment

ON Student.StudentID = Enrollment.StudentID;



Q6) Write a MySQL query using RIGHT JOIN to display all courses, including courses that have no enrolled students.

The output should include:

Column
StudentName
CourseID
CourseName
Fees

SELECT

Student.StudentName,

Course.CourseID,

Course.CourseName,

Course.Fees

FROM Enrollment

RIGHT JOIN Course

ON Enrollment.CourseID = Course.CourseID

LEFT JOIN Student

ON Enrollment.StudentID = Student.StudentID;

The screenshot displays the MySQL Workbench interface. The central editor shows a SQL query in 'SQL File 7' with the following code:

```
4 Student.StudentName,  
5 Course.CourseID,  
6 Course.CourseName,  
7 Course.Fees  
8 FROM Enrollment  
9 RIGHT JOIN Course  
10 ON Enrollment.CourseID = Course.CourseID  
11 LEFT JOIN Student  
12 ON Enrollment.StudentID = Student.StudentID;
```

Below the query editor, the 'Result Grid' shows the output of the query. The columns are StudentName, CourseID, CourseName, and Fees. The data is as follows:

StudentName	CourseID	CourseName	Fees
Rahul Sharma	101	BSc IT	45000.00
Sneha Joshi	101	BSc IT	45000.00
Priya Patil	102	BCA	40000.00
Karan Mehta	102	BCA	40000.00
Aman Verma	103	BSc CS	42000.00

On the left, the 'SCHEMAS' pane shows the 'collegedb' database with tables 'course', 'enrollment', and 'student'. The bottom status bar indicates the system time as 05:01 PM on 16-08-2026.